

Mathematical Analysis II

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Contents

Preface	iv
1 Sequences and Series of Functions	1
1.1 Discussion of Main Problem	1
1.2 Uniform Convergence	1
1.3 Uniform Convergence and Continuity	1
1.4 Uniform Convergence and Integration	1
1.5 Uniform Convergence and Differentiation	1
1.6 Equicontinuous Families of Functions	1
1.7 The Stone-Weierstrass Theorem	1
2 Some Special Functions	2
2.1 Power Series	2
2.2 The Exponential and Logarithmic Functions	2
2.3 The Trigonometric Functions	2
2.4 The Algebraic Completeness of the Complex Field	2
2.5 Fourier Series	2
2.6 The Gamma Function	2
3 Functions of Several Variables	3
3.1 Differentiation	3
3.2 The Contraction Principle	3
3.3 The Inverse Function Theorem	3
3.4 The Implicit Function Theorem	3
3.5 The Rank Theorem	3
3.6 Derivatives of Higher Order	3
3.7 Differentiation of Integrals	3
4 Integration of Differential Forms	4
4.1 Integration	4
4.2 Primitive Mappings	4
4.3 Partitions of Unity	4
4.4 Change of Variables	4

4.5	Differential Forms	4
4.6	Simplexes and Chains	4
4.7	Stokes' Theorem	4
4.8	Closed Forms and Exact Forms	4
4.9	Vector Analysis	4
5	The Lebesgue Theory	5
5.1	Set Functions	5
5.2	Construction of the Lebesgue Measure	5
5.3	Measure Spaces	5
5.4	Measurable Functions	5
5.5	Simple Functions	5
5.6	Integration	5
5.7	Comparison with the Riemann Integral	5
5.8	Integration of Complex Functions	5
5.9	Functions of Class $\mathcal{L}^2$	5

Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.33.1 and Mathlib v4.33.1 (revision [0df444a360ea](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.

Chapter 1

Sequences and Series of Functions

1.1 Discussion of Main Problem

1.2 Uniform Convergence

1.3 Uniform Convergence and Continuity

1.4 Uniform Convergence and Integration

1.5 Uniform Convergence and Differentiation

1.6 Equicontinuous Families of Functions

1.7 The Stone-Weierstrass Theorem

Chapter 2

Some Special Functions

2.1 Power Series

2.2 The Exponential and Logarithmic Functions

2.3 The Trigonometric Functions

2.4 The Algebraic Completeness of the Complex Field

2.5 Fourier Series

2.6 The Gamma Function

Chapter 3

Functions of Several Variables

3.1 Differentiation

3.2 The Contraction Principle

3.3 The Inverse Function Theorem

3.4 The Implicit Function Theorem

3.5 The Rank Theorem

3.6 Derivatives of Higher Order

3.7 Differentiation of Integrals

Chapter 4

Integration of Differential Forms

4.1 Integration

4.2 Primitive Mappings

4.3 Partitions of Unity

4.4 Change of Variables

4.5 Differential Forms

4.6 Simplexes and Chains

4.7 Stokes' Theorem

4.8 Closed Forms and Exact Forms

4.9 Vector Analysis

Chapter 5

The Lebesgue Theory

5.1 Set Functions

5.2 Construction of the Lebesgue Measure

5.3 Measure Spaces

5.4 Measurable Functions

5.5 Simple Functions

5.6 Integration

5.7 Comparison with the Riemann Integral

5.8 Integration of Complex Functions

5.9 Functions of Class $\mathcal{L}^2$

Bibliography

- [1] Walter Rudin, *Principles of Mathematical Analysis*, McGraw-Hill International Editions.